



$[1, 2, 3, 4, 5, 6]$, $K=4$ th ✓
 1 2 3 4 5 6

$-1 -1 -1 1 2 3$
 ↑ array

for large

min
 heap ✓

2	7	3	6	8	11	4	9	18
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↑ ↑ ↑ ↑ ↑
 K=4

$[18, 9, 8, 11]$ ← min heap
 possible window

$-1 -1 -1 2 6 6 7 8$

To 3 small

Sum1 $\Rightarrow 68 - 22 = 46$

24
-12
+10

$\Rightarrow 50 - 22 = 28$
+1
39
-20
+12

Sum2 = 0

68
-22
+14

20, 8, 14, 4, 12, 10

Max $\rightarrow$ heap

22

[20, 8, 22, 4, 12, 10, 14]

$K1 = 3, K2 = 6$

$K1^{th}$ smallest integer

$K2^{th}$ smallest integer

4, 8, 10, 12, 14, 20, 22

Small $\rightarrow A$

$B - A - 6^{th}$ small

Sum = B

max heap

[10, 5, 6, 2] $\leftarrow n$

[12, 7, 9] $\leftarrow m$

max heap

single max heap

array

[10, 5, 6, 2, 12, 7, 9]

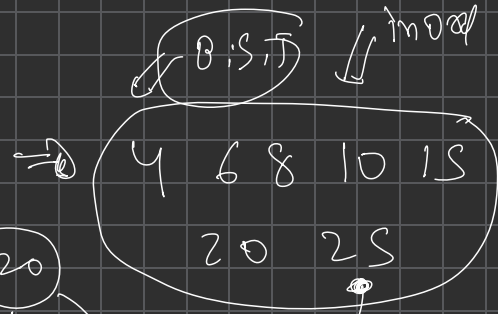
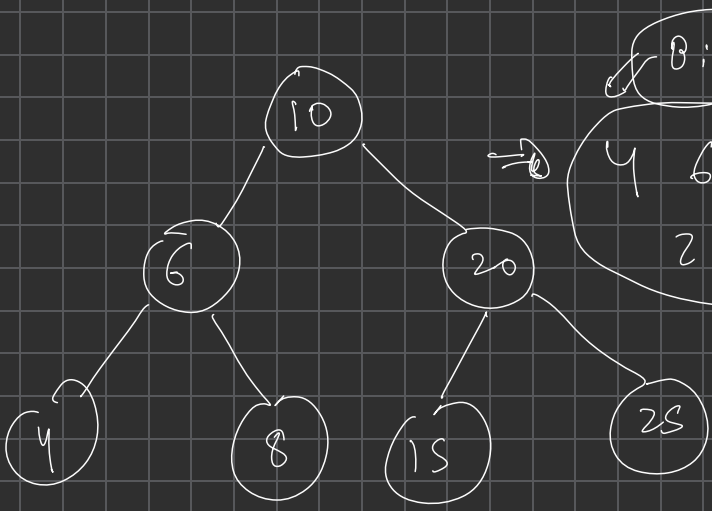
Step down

$n+m$

$\Theta(n+m)$

12 10 9 7 6 5 2

50 R2
←



2 R 10
post order

ma

